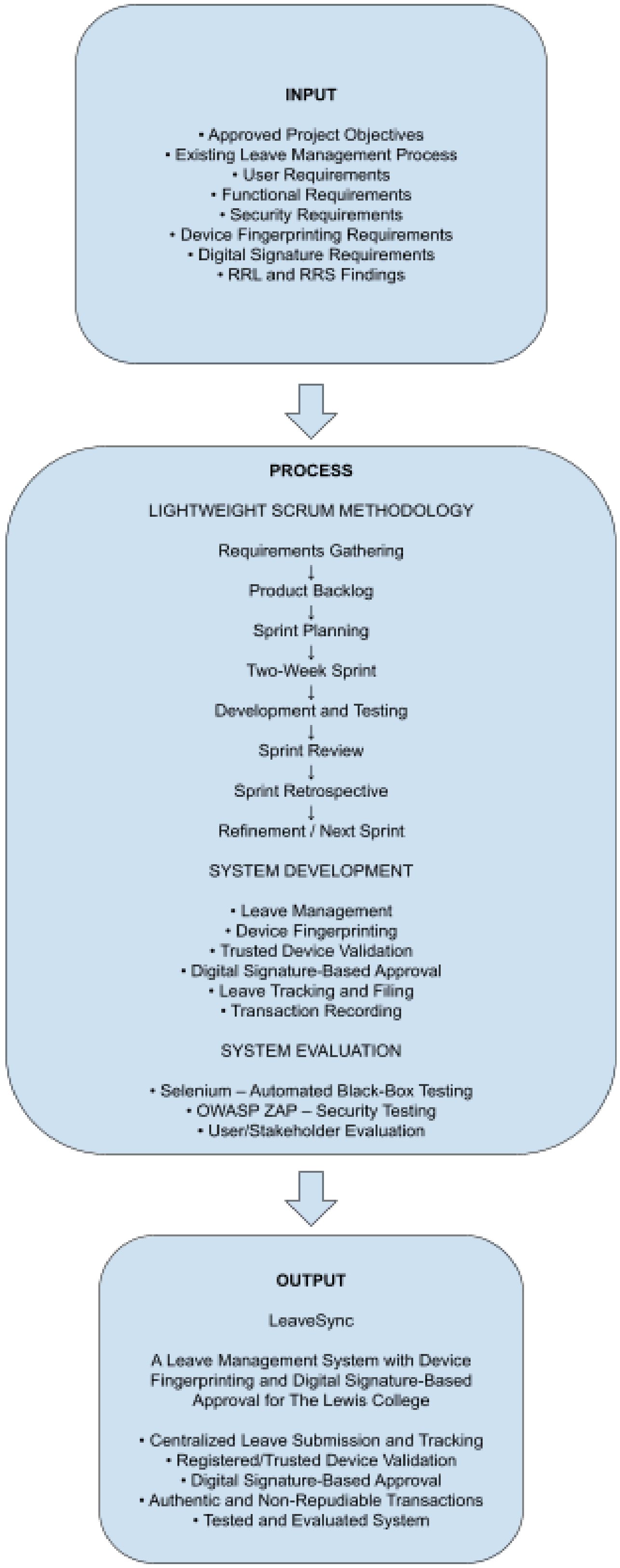
# CHAPTER 3: PROJECT METHODOLOGY

## Conceptual Framework

The conceptual framework of the study uses the **Input-Process-Output (IPO) Model**. It illustrates how the requirements and resources identified for LeaveSync will be transformed through the **Lightweight Scrum Methodology**, system development activities, and system evaluation into the proposed leave management system.

## Conceptual Framework Diagram



## Description of the Conceptual Framework

The **Input** consists of the information and requirements needed for the development of LeaveSync. These include the approved objectives, existing leave management procedures, employee and user requirements, device fingerprinting requirements, digital signature requirements, and findings from the Review of Related Literature and Related Studies. The device fingerprinting component will consider browser type, operating system version, and IP patterns, as specified in the approved project objectives.

The digital signature requirements focus on cryptographic signing, verification of the identity of the authorized manager or approver, authenticity, and non-repudiation of approved transactions.

The **Process** will use the **Lightweight Scrum Methodology**. The methodology follows an iterative development approach in which requirements are gathered, organized into a Product Backlog, prioritized, and developed through two-week Sprints. Each Sprint will include planning, development, testing, review, and retrospective activities. The latest Scrum lesson specifies the use of a two-week Sprint, Sprint Planning, Daily Check-ins, Sprint Review, Sprint Retrospective, and a task board for monitoring development progress.

The system development process will focus on the web-based leave management module, device fingerprinting, trusted-device validation, digital signature-based approval, leave tracking and filing, and transaction recording.

The system will be evaluated through **Selenium automated black-box testing** and **OWASP ZAP security testing**, as required by the fourth objective of the study. User or stakeholder evaluation will also be conducted to obtain feedback regarding the developed system.

The **Output** is **LeaveSync: A Leave Management System with Device Fingerprinting and Digital Signature-Based Approval for The Lewis College**. The system is intended to provide centralized leave submission and tracking while incorporating device-aware security and cryptographic approval mechanisms.

The framework follows the traceability principle presented in the Lightweight Scrum lesson:

**Requirement → User Story → Sprint Task → Working Increment → Test Result → Stakeholder Acceptance.**

# Project Methodology

The researchers will use the **Lightweight Scrum Methodology** in developing LeaveSync. Lightweight Scrum is appropriate for the project because the development team consists of three members and the proposed system contains several interconnected features that require incremental development, continuous testing, and regular evaluation.

The system will be developed through **two-week Sprints**. Each Sprint will have a specific goal and will produce a working product increment. The development process will follow requirements gathering, Product Backlog creation, Sprint Planning, development, testing, Sprint Review, and Sprint Retrospective activities.

## Scrum Team and Responsibilities

The three researchers will form the Scrum development team. Because the project uses a lightweight implementation of Scrum, the responsibilities will be distributed among the researchers while allowing members to perform multiple functions.

### Product Owner

The Product Owner will represent the project requirements and prioritize the Product Backlog. The Product Owner will help determine which features and requirements should be developed first and will participate in reviewing and accepting completed product increments.

The Product Owner may also contribute to development activities as part of the three-person team.

### Scrum Facilitator

One researcher will serve as the Scrum Facilitator. The facilitator will coordinate Scrum activities, assist in organizing meetings, monitor Sprint progress, and help identify and resolve development blockers.

The Scrum Facilitator may also participate in development, testing, and documentation activities.

### Developers

The three researchers will collaborate in development activities. Their responsibilities will include requirements analysis, interface design, programming, database development, testing, documentation, and system refinement.

The researchers will not strictly separate their development responsibilities because the Lightweight Scrum approach is intended for a small development team where members collaborate on analysis, coding, testing, and documentation.

## Requirements Gathering

The researchers will gather and analyze the requirements necessary for the development of LeaveSync.

The activities will include:

1. Reviewing the approved project objectives;
2. Reviewing the existing leave application and approval process;
3. Identifying employee requirements;
4. Identifying administrator and authorized approver requirements;
5. Identifying leave submission and tracking requirements;
6. Identifying device fingerprinting requirements;
7. Identifying digital signature requirements;
8. Identifying functional and security requirements; and
9. Reviewing the RRL and RRS for relevant findings that support the proposed system.

The gathered requirements will be translated into **user stories** and added to the Product Backlog.

## Product Backlog

The Product Backlog will contain the prioritized requirements of LeaveSync in the form of user stories. It will be continuously refined throughout the development process.

The initial Product Backlog will include the following:

| ID | User Story |
|---|---|
| US-01 | As an employee, I want to log in so that I can securely access the system. |
| US-02 | As an employee, I want to register my device so that I can use a trusted device for leave submission. |
| US-03 | As an employee, I want the system to verify my device so that unauthorized devices cannot submit leave applications. |
| US-04 | As an employee, I want to submit a leave request online so that I do not need to use a manual process. |

| | |
|---|---|
| US-05 | As an employee, I want to track my leave request so that I can monitor its status. |
| US-06 | As an authorized approver, I want to view leave requests so that I can evaluate them. |
| US-07 | As an authorized approver, I want to digitally sign an approval so that the transaction is authenticated and non-repudiable. |
| US-08 | As an administrator, I want to manage users and registered devices so that system access can be controlled. |
| US-09 | As an administrator, I want to view leave records so that leave information can be monitored and maintained. |
| US-10 | As a system administrator, I want the system to maintain transaction records so that leave activities can be traced. |

The Product Backlog will be reviewed and prioritized during Sprint Planning. Items not selected for the current Sprint will remain in the Product Backlog for future Sprints.

## Sprint Plan

The researchers will use **two-week Sprints** throughout the development of LeaveSync.

| Sprint | Duration | Sprint Goal | Expected Increment |
|---|---|---|---|
| Sprint 1 | 2 weeks | Establish the system foundation | User registration, login, roles, and database foundation |
| Sprint 2 | 2 weeks | Develop core leave management | Leave submission, tracking, and status management |
| Sprint 3 | 2 weeks | Implement device security | Device registration, fingerprint generation, and validation |
| Sprint 4 | 2 weeks | Implement secure approval | Approval workflow and cryptographic digital signature |
| Sprint 5 | 2 weeks | Integrate and test the system | Integrated system, Selenium testing, and OWASP ZAP testing |
| Sprint 6 | 2 weeks | Refine and finalize the system | Corrections, improvements, and final tested system |

Each Sprint will begin with **Sprint Planning**. During development, the researchers will conduct short **Daily Check-ins** to monitor progress and identify impediments.

At the end of each Sprint, the researchers will conduct a **Sprint Review** to demonstrate the completed increment and gather feedback. A **Sprint Retrospective** will then be conducted to identify improvements that can be applied to the succeeding Sprint.

# Work Breakdown Structure

The project tasks will be divided into manageable activities through a Work Breakdown Structure.

## 1. Project Preparation

- Review approved objectives
- Review RRL and RRS
- Gather system requirements
- Identify stakeholders
- Develop user stories

## 2. System Design

- Design system architecture
- Design database
- Design user interface
- Design leave workflow
- Design device fingerprinting workflow
- Design digital signature workflow

## 3. Leave Management Development

- Develop authentication
- Develop user roles
- Develop leave application
- Develop leave tracking
- Develop leave filing
- Develop leave records

## 4. Device Fingerprinting Development

- Develop device registration
- Collect browser information
- Collect operating system information
- Collect IP pattern information

- Generate device fingerprint
- Compare fingerprint with registered device
- Restrict unauthorized devices

### 5. Digital Signature Development

- Develop approver authentication
- Develop approval workflow
- Implement cryptographic signing
- Implement signature verification
- Record approval transactions

### 6. System Integration

- Integrate leave management
- Integrate device validation
- Integrate digital approval
- Integrate transaction records

### 7. Testing

- Prepare Selenium test cases
- Conduct automated black-box testing
- Conduct OWASP ZAP security testing
- Identify defects
- Correct defects
- Conduct retesting

### 8. Evaluation and Finalization

- Conduct user evaluation
- Conduct Sprint Review
- Conduct Sprint Retrospective
- Implement improvements
- Complete documentation
- Prepare final system demonstration

## Increment and Sprint Review Results

At the end of every Sprint, the researchers will document the completed product increment and the results of the Sprint Review.

| Sprint | Sprint Goal | Features Completed | Review/Feedback | Action Taken |
|---|---|---|---|---|
| Sprint 1 | Establish system foundation | To be completed during Sprint | To be documented | To be documented |
| Sprint 2 | Develop leave management | To be completed during Sprint | To be documented | To be documented |
| Sprint 3 | Implement device security | To be completed during Sprint | To be documented | To be documented |
| Sprint 4 | Implement digital approval | To be completed during Sprint | To be documented | To be documented |
| Sprint 5 | Integrate and test | To be completed during Sprint | To be documented | To be documented |
| Sprint 6 | Refine and finalize | To be completed during Sprint | To be documented | To be documented |

The completed increment at the end of each Sprint will represent the functionality completed during that Sprint. The Sprint Review will provide an opportunity for the researchers and stakeholders to examine the increment and provide feedback.

Feedback obtained during the Sprint Review will be considered in the refinement and prioritization of the Product Backlog for succeeding Sprints.

# Testing and Acceptance Criteria

Testing will be conducted throughout the development process to ensure that each completed increment meets its intended requirements.

## Automated Black-Box Testing

**Selenium** will be used for automated black-box testing of the system.

The test cases will cover major system functions, including:

- User login;
- Device registration;
- Device verification;
- Leave submission;
- Leave tracking;
- Unauthorized-device restriction;

- Leave approval;
- Digital signature execution;
- Signature verification; and
- Transaction recording.

A test case will be considered successful when the actual system behavior corresponds to the expected result.

## Security Testing

**OWASP ZAP** will be used to conduct security testing of the LeaveSync web application.

The testing will examine potential security weaknesses involving:

- Authentication;
- Access control;
- Session management;
- Input handling;
- Web application vulnerabilities; and
- Security of system endpoints.

Identified vulnerabilities will be documented and addressed by the researchers. Corrective changes will be followed by retesting.

## User Acceptance

After functional and security testing, the completed system will be presented to the selected project respondents.

The respondents will evaluate the developed system based on the established evaluation criteria. Their feedback will be used to determine whether the system satisfies the intended requirements and to identify areas for improvement.

The system will be considered ready for final acceptance when:

1. The required system features have been implemented;
2. The major Selenium test cases produce the expected results;
3. Critical security issues identified through OWASP ZAP have been addressed;
4. Registered devices are properly validated;
5. Unauthorized devices are appropriately restricted from submitting leave applications;
6. The digital signature mechanism properly authenticates approval transactions;
7. Major defects identified during testing have been corrected; and
8. The system receives acceptance from the appropriate stakeholders.

The testing process will maintain traceability from the requirement through the user story, Sprint task, working increment, test result, and stakeholder acceptance.

# Project Respondents

The project respondents will consist of **at least 15 individuals** who are relevant to the use and evaluation of LeaveSync at The Lewis College.

The researchers will use **purposive sampling** to select the respondents because the evaluation requires individuals who can provide relevant feedback regarding the proposed leave management process.

The respondents will include potential employee users who will evaluate the leave submission, tracking, and filing functions, as well as authorized personnel who will evaluate the leave approval, device validation, and digital signature features.

A minimum of **15 respondents** will participate in the evaluation of the developed system. Their feedback will be used to determine the acceptability, usability, functionality, and security-related aspects of LeaveSync and to identify areas requiring improvement.

# Programming Language and Software Used

The following programming languages and software will be used in the development, testing, and implementation of LeaveSync.

## HTML5

HTML5 will be used to structure the web pages, forms, dashboards, tables, and navigation components of the system.

## CSS3

CSS3 will be used to design the appearance and layout of the system, including spacing, typography, responsiveness, and other interface elements.

## JavaScript

JavaScript will be used for client-side interactivity, form validation, asynchronous functions, and browser-based processes required for device fingerprinting.

## PHP

PHP will be used for server-side programming and business logic. It will process leave requests, manage authentication and authorization, handle approval transactions, and communicate with the database.

## MySQL

MySQL will be used as the relational database management system for storing employee accounts, leave applications, device information, approval records, and other system data.

## Visual Studio Code

Visual Studio Code will be used as the primary development environment for writing, editing, debugging, and organizing the source code.

## XAMPP

XAMPP will be used as the local web-server and database development environment.

### Git

Git will be used for source-code version control and tracking changes throughout development.

### GitHub

GitHub will be used as the remote repository for source-code storage, version management, and collaboration.

### Figma

Figma will be used to create wireframes, prototypes, and user-interface designs before and during system development.

### Selenium

Selenium will be used to conduct automated black-box testing of the web application.

### OWASP ZAP

OWASP ZAP will be used for security testing and vulnerability assessment of the developed web application.

### Cryptographic Signing Technology

A cryptographic signing mechanism will be used for the digital signature-based approval component. The final implementation will use an appropriate cryptographic approach, such as **WebAuthn or private-key-based signing**, based on the final technical design.

The digital signature mechanism will be used to authenticate authorized approvals and provide transaction authenticity and non-repudiation.

# Documentation Tools

The following tools will be used for the documentation and management of the project.

### Google Docs

Google Docs will be used for collaborative preparation, editing, and review of research and project documents.

### Figma

Figma will be used to document wireframes, prototypes, user-interface designs, and screen layouts.

### GitHub

GitHub will be used to maintain the project's source-code repository and development history.

### draw.io / diagrams.net

draw.io or diagrams.net will be used to create flowcharts, system diagrams, database diagrams, and other graphical representations.

### Trello / Task Board

Trello or a similar task-board tool will be used to manage the Product Backlog, Sprint tasks, and development progress.

The task board will follow the workflow:

**Backlog → To Do → In Progress → Testing/Review → Done**

This allows the researchers to monitor the status of development tasks throughout each Sprint.

### Selenium

Selenium will be used to document automated test cases, test scenarios, expected results, actual results, and test outcomes.

### OWASP ZAP

OWASP ZAP will be used to generate and document security testing results, including identified vulnerabilities and corrective actions.

# Project Cost

The project cost represents the expenses incurred by the researchers during the development, testing, documentation, and presentation of LeaveSync. Since the researchers are students with limited financial resources, available personal resources and free or open-source software will be utilized whenever possible.

| No. | Expense/Resource | Description | Estimated Cost |
|---|---|---|---|
| 1 | Internet Connection | Research, development, communication, and system testing | ₱1,500.00 |
| 2 | Transportation | Consultations, data gathering, and project-related activities | ₱1,000.00 |
| 3 | Printing | Questionnaires, testing documents, and other required materials | ₱500.00 |
| 4 | Presentation Materials | Materials for system demonstration and presentation | ₱300.00 |
| 5 | Hosting/Deployment | Basic hosting or deployment expenses, if required | ₱500.00 |
| 6 | Miscellaneous | Other minor project-related expenses | ₱500.00 |
| 7 | Software Development Value | Estimated value of the completed LeaveSync system | ₱5,000.00 |
| | **Total Estimated Project Cost** | | **₱9,300.00** |